

GROWTH HORMONE AND PROLACTIN RESPONSE TO METHYLPHENIDATE IN
CHILDREN WITH ATTENTION DEFICIT DISORDER

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Summary

Utilizing a double-blind, drug-placebo design, we examined growth hormone (GH) and prolactin (Pro) response to oral administration of methylphenidate (MPH) in 14 boys (ages 7.0-12.4 years) with Attention Deficit Disorder (ADD). Four conditions representing three different MPH doses (0.3 mg/kg O.D., 0.3 mg/kg B.I.D., 0.6 mg/kg O.D.) and Placebo were compared in each subject, each condition lasting for a period of 3 weeks. GH and Pro response were measured both as maximum peak GH (DGH) or nadir of Pro (DPro) as well as area under the curve for the first four hours after MPH administration (AUCGH, AUCPro). Behavioral measures included parent ratings on the Yale Children's Inventory and teacher ratings on the Yale and Conners Behavior Rating Scales and Kagan's Matching Familiar Figures Test (MFFT). Prolactin response as measured by AUCPro was significantly increased after MPH compared to placebo ($t=2.04$, $p<0.05$, placebo vs all doses MPH). This difference observed for AUCPro between placebo and MPH was evident as well when we considered the number of times AUCPro declined after MPH as compared to placebo ($p=.018$, Fisher's exact test). Within-subjects analysis of covariance demonstrated significant correlations between the improvement in reaction time on the MFFT and 1) GH response (AUCGH, $r=.58$, $p<.001$) and 2) prolactin response (AUCPro, $r=.40$, $p<.05$) and between improvement in attention as measured on the Yale BRS and GH response (AUCGH, $r=.57$, $p<.05$). Our findings suggest that measures other than GH and prolactin may be more desirable measures of brain catecholaminergic functioning.

Good evidence from many laboratories has accumulated to support the belief that methylphenidate (MPH) and its closely related phenethylamine derivative, amphetamine, often produce remarkable ameliorative effects on attention modulation and ac-

tivity regulation in children with ADD (1-3). Pharmacological studies in both animals and man have demonstrated that the stimulants exert their actions via central catecholaminergic mechanisms (4,5). Thus, like amphetamine, MPH exerts its central actions by facilitating the release of catecholamines from nerve terminals in brain and by inhibiting their reuptake. In contrast to amphetamine which appears to facilitate the release of amine from a newly synthesized pool, MPH represents the prototype of drugs whose actions are blocked by reserpine and, thus, presumably releases catecholamine from a storage pool (4,6). These actions, coupled with the proven safety of the agent, suggest that MPH would provide a safe and theoretically rational pharmacologic probe to perturb central monoaminergic systems in children with ADD.

This investigation utilized the pharmacologic probe methodology to examine the role of monoaminergic mechanisms in boys with Attention Deficit Disorder. The general principle involves the administration of a particular pharmacologic agent which is designed to affect specific neural systems. Its effects are determined by using as dependent measures the concentrations of compounds in plasma and urine that reflect the activity of the neural system under investigation. In this report we detail our findings employing methylphenidate as the pharmacologic probe and the determinations of plasma growth hormone and prolactin as indices of central monoaminergic systems. Our rationale was derived from the not inconsiderable number of psychoneuroendocrinologic investigations which indicate that excitatory dopaminergic, noradrenergic and serotonergic mechanisms are involved in growth hormone release while prolactin appears to be under the control of both inhibitory dopaminergic and excitatory serotonergic mechanisms (7-9).

Methods

A. Subject Selection

Our patient population was comprised of 14 boys (7.0-12.4 years, mean 10.4 ± 0.51 , mean $\pm$ SEM) satisfying "best estimate" (10) diagnostic criteria for ADD. The subjects were selected from three referral sources: the Pediatric Neurology Clinic and the Learning Disorders Unit of the Yale New Haven Hospital; and Yale Child Study Center clinics. Their scores (mean $\pm$ SEM) on the Wechsler Intelligence Scale for Children-Revised (WISC-R, $n=13$) or Stanford-Binet ($n=1$) averaged: Verbal, 97.9 ± 2.82 ; Performance, 104 ± 3.62 and Full Scale, 100 ± 2.72 . Socioeconomic status, reflected in the Hollingshead index, was uniform within the group, with all subjects exhibiting scores between 2-3, reflecting a predominantly middle-class environment. All boys were in Tanner stage 2.

B. Experimental Protocol

The experiment consisted of a double-blind, drug-placebo study in which each subject served as his own control. Every subject participated in four medication conditions, each of which lasted for a period of three weeks: methylphenidate (MPH) at one of three doses (0.3 mg/kg OD; 0.3 mg/kg BID; 0.6 mg/kg OD) and a placebo (BID). During the third week of each condition the children were admitted to the Children's Clinical Research Center (CCRC) at 7:30 am. MPH or placebo was administered at 9 am; the second

dose (for 0.3 mg/kg BID and placebo conditions) was administered at 1 pm. An intravenous line was placed on admission at 7:30 a.m. The first blood sample, a baseline sample, was drawn at 8:30 and a second baseline sample at 8:50. Blood samples were collected every 30 minutes for the first three hours and once an hour thereafter for a total of 10-12 hours. On the following day, the same dose of MPH (or placebo) was administered, but no intravenous line was placed. Attentional tests were administered one hour prior to MPH and two hours after MPH. Plasma was kept on ice and analyzed within two weeks for growth hormone and prolactin. Growth hormone was measured using a modification of the radioimmunoassay of Rosselyn et al. (11) and prolactin levels were determined by a modification of the radioimmunoassay method of Sinha et al. (12). All children received a standard breakfast at 7:45 and lunch at 12 noon, composed of low monoamine, low xanthine diet, providing 14.5 gm protein, 11 gm of carbohydrates and 13.2 gm of lipid.

C. Diagnostic Instruments

Parents completed the Yale Children's Inventory (13,14) and the 39-item Conners' Parent-Teacher Scale (15) prior to beginning MPH, and teachers completed both the Conners' Abbreviated Teacher Questionnaire (15) and the Yale Teacher Behavior Rating Scale (BRS) at weekly intervals. The Matching Familiar Figures Test (MFFT, 16) served as the principal measure of attention-impulsivity. Four different forms of the MFFT were used in this study so that each form did not have to be repeated in any subject.

Results

Previous studies by our group (17) indicated that baseline concentrations of growth hormone and prolactin appear to be normal in children with ADD. A principal focus of the present investigation was to determine the growth hormone and prolactin response following administration of MPH. We scored the change in growth hormone from baseline to peak, termed the delta growth hormone (DGH), or when considering prolactin, the change from baseline to its nadir, termed delta prolactin (DPro). We also determined the total area under the curve for the first four hours after MPH administration and termed area under the curve for growth hormone (AUCGH) and area under the curve for prolactin (AUCPro), respectively. These were calculated by applying the trapezoidal rule in a short subroutine written for an Apple II Plus computer (D. Shaywitz). AUCGH represents the algebraic sum of all values above baseline minus those values below baseline. AUCPro was calculated in a similar fashion, except that values below baseline were expected, and thus, values above baseline were subtracted from the sum of those below. Neuroendocrine (DGH, DPro, AUCGH and AUCPro) and behavioral (scales on the Yale Children's Inventory, Conners, and Yale Behavior Rating Scales) and Matching Familiar Figures test results) variables were compared for the four conditions using Statistical Analysis System (SAS) procedures. There were no significant changes among the four conditions for DGH, AUCGH, or DPro. However, as shown in FIG. 1, AUCPro was significantly increased after MPH compared to placebo ($t=2.04$, $p<0.05$, placebo vs all doses MPH).

This difference observed for AUCPro between placebo and MPH was evident as well when we considered the number of times AUCPro declined after MPH as compared to placebo ($p=.018$, Fisher's exact

test). There was no significant difference between the means for the three MPH conditions $F=1.77$, $p>.1$), although the highest dose of MPH resulted in the largest mean AUCPro (FIG. 1).

The relationship between AUCGH X DGH and AUCPro X DPro were significant for all conditions (TABLE I). In general, there was poor correlation for either growth hormone or prolactin response between condition 2 (MPH 0.3 mg/kg at 9 a.m.) and condition 3 (MPH 0.3 mg/kg at 9 a.m. and 1 p.m.) which should have been similar for the first four hours of the day. We did not find any consistent correlation between growth hormone and prolactin response for any of the conditions. Significant correlations emerged between DPro and the teachers rating of hyperactivity and the Yale Behavioral Rating Scale ($r=.76$, $p<.001$).

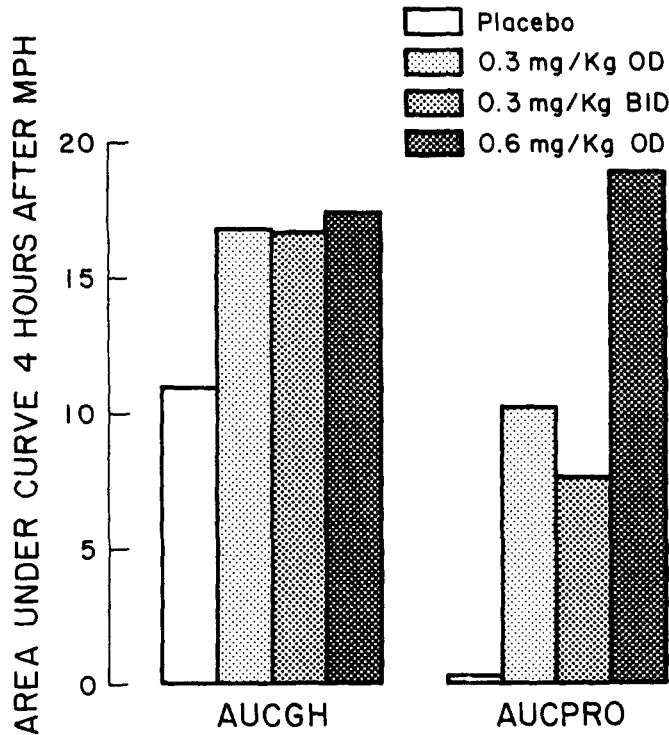


FIG. 1

Mean values for AUCGH and AUCPro in each of four conditions of the study. Mean AUCPro in the placebo condition was significantly less than for the methylphenidate conditions. Although the values did not achieve significance there was a trend for the highest AUCPro to be found at the maximum dose of methylphenidate. No difference was noted for DGH, AUCGH or DPro between conditions.

TABLE I
Relationship Between Neuroendocrine Measures

Variable	r	p
DGH1 X AUCGH1	.81	.008
DGH2 X AUCGH2	.80	.002
DGH3 X AUCGH3	.86	.003
DGH4 X AUCGH4	.86	.001
DPro1 X AUCPro1	.68	.06
DPro2 X AUCPro2	.94	.001
DPro3 X AUCPro3	.88	.002
DPro4 X AUCPro4	.87	.004

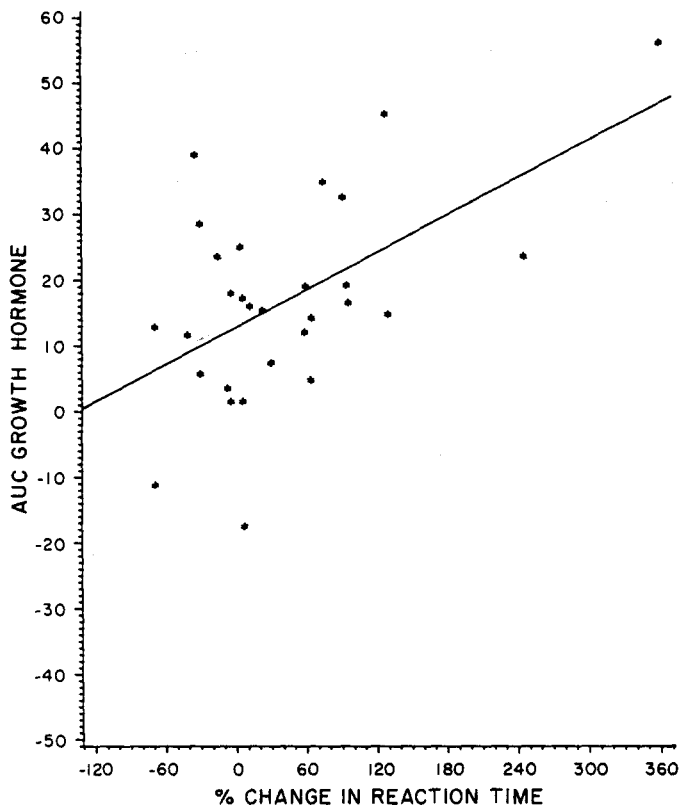


FIG. 2

Relationship between growth hormone response to methylphenidate as measured by AUCGH and the percent improvement as measured by the reaction time on the MFFT. All methylphenidate conditions are combined in this analysis. The greater the growth hormone response to methylphenidate, the greater the improvement in reaction time ($r=.58$, $p<.001$).

Our finding that an increase in plasma GH as measured by the AUCGH was inversely related to the number of errors on the MFFT (FIG. 2) is difficult to reconcile since we did not find any consistent relationship between GH measures (either DGH or AUCGH) and MPH.

TABLE II

Relationship Between Behavioral Variables

Variable	r	p
activity X tractability	.77	.003
activity X CD-socialized	.63	.02
attention X language	.65	.02
impulsivity X CD-aggressive	.70	.01
tractability X CD-socialized	.66	.02
tractability X state	.65	.02

Significant correlations were found among behavioral variables (TABLE II), for example, YCI scales of activity and tractability, activity and conduct disorder-socialized type; attention and language; impulsivity and conduct disorder-aggressive type; tractability and state (comprising primarily measures of sleep) and tractability and conduct disorder, socialized type. We also noted significant correlations between particular YCI scale scores and measures on the MFFT, most consistently with the anxiety-emotional scale.

Discussion

In a previous study utilizing an open trial of MPH at a dose of 0.34 mg/kg (17) we noted significant increases in GH concentrations from a mean of 4.4 ng/ml at baseline to peak at 10.5 at two hours after MPH administration. Such findings are comparable to those reported by Aarskog et al. (18) who compared the GH response for the first three hours after drug administration between oral L-DOPA, amphetamine and methylphenidate in hyperactive children 6-13 years of age. An increase in GH from baseline levels of 3 ng/ml to 10 ng/ml was demonstrated both acutely and after chronic methylphenidate administration. This increase in GH has been found in adults as well. Thus, Brown et al. (19), in a study of 59 healthy young men, reported significant increases in GH after 20 mg amphetamine or 20 mg MPH. Similar findings have been noted also by others (20-22).

In our previous study (17) we noted a decline in prolactin from initial concentrations of 9.5 ng/ml to a nadir of 3.8 ng/ml 1.5 hours after MPH. Prolactin decline following stimulants have been reported by Wells et al. (23) who studied 6 young adults and noted significant reductions in prolactin after 10 or 20 mg of amphetamine, findings reported as well by others (19, 24). However, most of these studies, including our own earlier report, failed to employ a double-blind, drug-placebo paradigm, a more rigorous methodology utilized in more recent studies. Thus, Gualtieri et al. (25), reported on 27 hyperactive children and 8 normal adults given MPH (0.3 and 0.6 mg/kg) or placebo. Growth hormone and prolactin were measured on just one occasion, one hour later. Mean growth hormone increased significantly after 0.3 mg/kg and still further after 0.6 mg/kg MPH. After 0.3 mg/kg, mean

prolactin did not decline significantly from placebo, but there was a significant, though minimal decline after 0.6 mg/kg.

Dommissse et al. (26) administered amphetamine or placebo to 6 men and 4 women, 19-37 years of age and measured growth hormone and prolactin 1, 1.5 and 2 hrs after. Following amphetamine, GH increased significantly from a baseline of 3.03 ng/ml to peak at 1.5 hours at 8.84 ng/ml, while little change was seen after placebo. In contrast, prolactin declined after both placebo and amphetamine and there was no significant difference between drug and placebo.

Garfinkel et al. (27) followed serial concentrations of GH and prolactin in 22 adolescents with histories of ADD as children (mean age 19 years) and 22 age matched controls after oral administration of amphetamine. Mean concentrations of GH increased from baseline levels of 1 ng/ml to peak at 10-15 ng/ml. Mean prolactin concentration decreased from baseline levels of 8-9 ng/ml to 4-5 ng/ml in both groups following amphetamine.

Weizman et al. (28) examined the neuroendocrinologic response to a methylphenidate challenge (5mg p.o.) on 16 school-aged children with ADD. Two conditions were employed: acute challenge and after 4 weeks of methylphenidate therapy. GH and prolactin were measured at baseline and two hours after methylphenidate administration. GH increased after the stimulant challenge in both the drug naive condition and after chronic administration. However, prolactin decreased in only the naive condition.

Our findings, utilizing a double-blind, drug-placebo methodology, suggest that the neuroendocrine response to MPH is not nearly as consistent in children with ADD as is reported in normal adults. Thus, we were unable to document any reliable increase in either of our two measures of GH (DGH or AUCGH) following MPH at either 0.3 or 0.6 mg/kg. Similarly, utilizing the change in prolactin concentration (DPro) we could not discern any reliable decrease in prolactin. However, when the AUCPro, the area under the curve for first four hours, was considered, significant reductions in prolactin after MPH were apparent. While there was a suggestion that higher doses of MPH produced greater alterations in AUCPro, this was not borne out statistically. Not surprisingly, the change in either GH (DGH) or Pro (DPro) was always significantly related to the AUCGH or AUCPro respectively.

Correlations with behavioral variables were few, though it appears that is a significant relationship between changes in GH following MPH and improvement in response time on the MFFT. However, in view of the lack of a consistent change in GH following MPH, the significance of this finding remains to be determined. The relationship noted between improvement in hyperactivity as measured by the Yale Behavior Rating Scale and reduction in prolactin as measured by the DPro is especially noteworthy in view of the narrow range that any changes in prolactin concentration must by necessity be measured. This narrow range produces a floor effect which tends to attenuate any significant effects, making the significant relationship particularly intriguing.

Our findings suggest that measures other than GH and prolactin would be more desirable measures of brain catecholaminergic functioning. Such measures might include actual determinations of

brain catecholamines or their metabolites in plasma and urine. Recent studies employing the monoamine oxidase inhibitor debrisoquin in conjunction with a pharmacological probe might provide a more useful index of central catecholaminergic mechanisms.

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